

ICD - Mechanical

XBANDigitalModule_20x20 - ICD Mechanical

Document Information

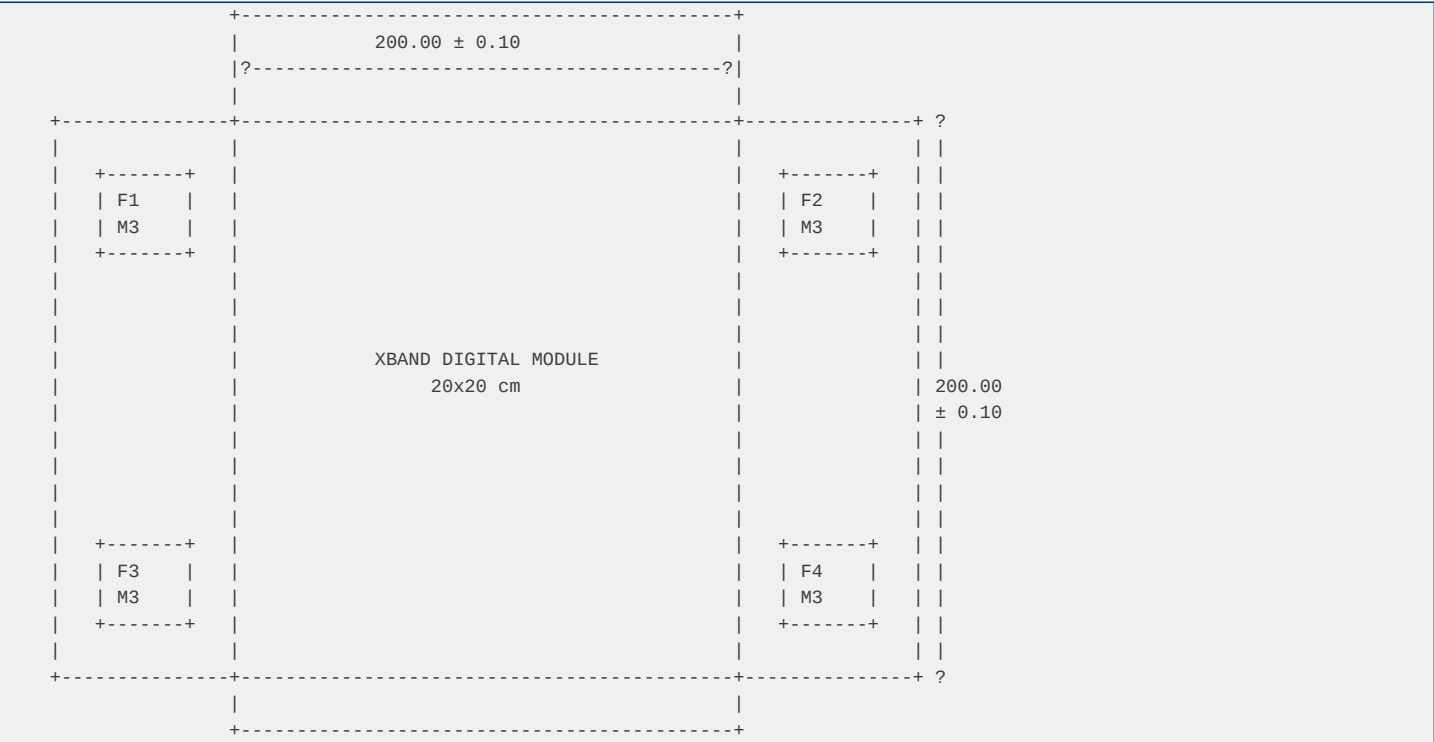
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Status	Preliminary Des

1. General Dimensions

1.1 Module Outline

Parameter	Nominal	Tolerance	Unit
Length	200.00	±0.10	mm
Width	200.00	±0.10	mm
Height (max)	25.00	±0.50	mm
PCB Thickness	2.40	±0.15	mm
Weight (max)	500	-	g

1.2 Mechanical Drawing Reference



2. Mounting Holes

2.1 Primary Mounting Holes (4x M3)

Hole	X Position	Y Position	Diameter	Type	Plating
F1	10.00 ± 0.05	10.00 ± 0.05	3.20 +0.10/-0.0	Through	Copper
F2	190.00 ± 0.05	10.00 ± 0.05	3.20 +0.10/-0.0	Through	Copper
F3	10.00 ± 0.05	190.00 ± 0.05	3.20 +0.10/-0.0	Through	Copper
F4	190.00 ± 0.05	190.00 ± 0.05	3.20 +0.10/-0.0	Through	Copper

2.2 Secondary Mounting Holes (2x M2.5)

Hole	X Position	Y Position	Diameter	Type	Plating
F5	100.00 ± 0.10	5.00 ± 0.10	2.70 +0.10/-0.0	Through	Copper
F6	100.00 ± 0.10	195.00 ± 0.10	2.70 +0.10/-0.0	Through	Copper

2.3 Mounting Hole Keep-Out

Hole	Keep-Out Radius	Layers	Notes
F1-F4	5.00 mm	All	No components,
F5-F6	4.00 mm	All	No components,

3. Connector Locations

3.1 RF Connector (SMA)

Parameter	Value	Tolerance
Type	SMA female, edg	-
Position X	5.00 mm	±0.10 mm
Position Y	100.00 mm	±0.10 mm
Orientation	Horizontal, lef	-
Impedance	50?	-
Frequency Range	DC - 18 GHz	-

3.2 Power Connector (MIL-DTL-38999)

Parameter	Value	Tolerance
Type	MIL-DTL-38999 S	-
Position X	100.00 mm	±0.10 mm
Position Y	5.00 mm	±0.10 mm
Orientation	Vertical, botto	-
Pin Count	4 (VIN, GND, GN	-
Current Rating	75A per contact	-

3.3 JTAG Connector

Parameter	Value	Tolerance
Type	Box header, 14-	-
Position X	195.00 mm	±0.10 mm
Position Y	180.00 mm	±0.10 mm
Orientation	Horizontal, rig	-
Pinout	Standard JTAG (	-

3.4 SpaceWire Connector

Parameter	Value	Tolerance
Type	MIL-DTL-38999 S	-
Position X	195.00 mm	±0.10 mm
Position Y	100.00 mm	±0.10 mm
Orientation	Vertical, right	-
Pin Count	8 (2 links: Dat	-

Data Rate		100 Mbps		-
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3.5 Debug Connector

Parameter		Value		Tolerance
Type		Box header, 10-		-
Position X		195.00 mm		±0.10 mm
Position Y		20.00 mm		±0.10 mm
Orientation		Horizontal, rig		-
Signals		UART (TX, RX),		-

4. Keep-Out Zones

4.1 RF Input Keep-Out

Parameter		Value
Area		X: 0-30 mm, Y:
Layers		All layers
Purpose		Isolate RF inpu
Components		Only RF compone
Traces		Only RF traces

4.2 Clock Keep-Out

Parameter		Value
Area		X: 170-200 mm,
Layers		All layers
Purpose		Isolate clock f
Components		Only clock comp
Traces		Only clock trac

4.3 Power Keep-Out

Parameter		Value
Area		X: 80-120 mm, Y
Layers		All layers
Purpose		Isolate power f
Components		Only power comp
Traces		Only power trac

4.4 FPGA Keep-Out

Parameter		Value
Area		X: 60-140 mm, Y
Layers		Top 4 layers (s
Purpose		FPGA core area
Components		Only FPGA and d
Traces		High-speed sign

5. PCB Stackup

5.1 Layer Structure (12 Layer)

Layer		Name		Thickness		Material		Purpose
1		Top Signal		0.035 mm		Copper		Component place
2		Ground 1		0.035 mm		Copper		Continuous grou
3		Signal 2		0.035 mm		Copper		Signal routing
4		Power 1		0.035 mm		Copper		1.0V core plane
5		Signal 3		0.035 mm		Copper		Signal routing

6		Ground 2		0.035 mm		Copper		Continuous grou
7		Ground 3		0.035 mm		Copper		Continuous grou
8		Power 2		0.035 mm		Copper		1.8V/3.3V plane
9		Signal 4		0.035 mm		Copper		Signal routing
10		Ground 4		0.035 mm		Copper		Continuous grou
11		Signal 5		0.035 mm		Copper		Signal routing
12		Bottom Signal		0.035 mm		Copper		Component place

5.2 Stackup Drawing

+-----+ Layer 1: Top Signal (35µm Cu) +-----+ Prepreg 2116 (0.10mm) +-----+ Layer 2: Ground 1 (35µm Cu) +-----+ Core (0.20mm) +-----+ Layer 3: Signal 2 (35µm Cu) +-----+ Prepreg 2116 (0.10mm) +-----+ Layer 4: Power 1 (35µm Cu) +-----+ Core (0.20mm) +-----+ Layer 5: Signal 3 (35µm Cu) +-----+ Prepreg 2116 (0.10mm) +-----+ Layer 6: Ground 2 (35µm Cu) +-----+ Core (0.20mm) +-----+ Layer 7: Ground 3 (35µm Cu) +-----+ Prepreg 2116 (0.10mm) +-----+ Layer 8: Power 2 (35µm Cu) +-----+ Core (0.20mm) +-----+ Layer 9: Signal 4 (35µm Cu) +-----+ Prepreg 2116 (0.10mm) +-----+ Layer 10: Ground 4 (35µm Cu) +-----+ Core (0.20mm) +-----+ Layer 11: Signal 5 (35µm Cu) +-----+ Prepreg 2116 (0.10mm) +-----+ Layer 12: Bottom Signal (35µm Cu) +-----+ Total Thickness: 2.40mm ± 0.15mm

5.3 Impedance Control

Trace Type	Impedance	Tolerance	Layer	Width
Single-ended 50	50?	±5%	1, 12	0.15 mm
Differential 10	100?	±5%	1, 12	0.12 mm (gap: 0
Single-ended 75	75?	±5%	3, 5, 9, 11	0.10 mm

6. Thermal Requirements

6.1 Thermal Interface

Parameter	Value
Material	Bergquist HI-60
Thickness	0.254 mm (10 mil)
Thermal Conduct	3 W/mK
Compression	10-30%

6.2 Thermal Via Pattern

Component	Via Diameter	Via Pitch	Pattern
FPGA BGA	0.30 mm	1.00 mm	Grid
ADC FCBGA	0.25 mm	0.80 mm	Grid
Power MOSFETs	0.30 mm	1.20 mm	Array

6.3 Heat Sink Requirements

Parameter	Value
Type	Passive, alumin
Height	5-10 mm
Fin Count	10-20 fins
Attachment	Thermal epoxy +
Thermal Resista	< 1.0 degC/W

7. Material Specifications

7.1 PCB Material

Parameter	Value
Base Material	Megtron6 (Mektr
Dielectric Cons	3.4 @ 1 GHz
Loss Tangent	0.002 @ 1 GHz
Tg	200 degC
CTE (x, y)	14 ppm/ degC
CTE (z)	45 ppm/ degC
Copper Weight	2 oz (70µm) on

7.2 Chassis Material

Parameter	Value
Material	Aluminum 6061-T
Finish	Hard anodize (T
Thickness	2.5 mm (base pl
Thermal Conduct	167 W/mK
CTE	23.6 ppm/ degC

7.3 Fastener Material

Parameter	Value
Material	Stainless Steel
Thread	M3 x 0.5, M2.5
Torque	M3: 0.5 N·m, M2
Locking	Nylon patch or

8. Environmental Requirements

8.1 Temperature

Parameter	Value
Operating	-40 degC to +85
Storage	-55 degC to +12
Thermal Shock	-55 degC to +12

8.2 Vibration

Parameter	Value
Random Vibratio	MIL-STD-810G, M
Frequency Range	20-2000 Hz
PSD	0.04 g²/Hz (20-
Duration	4 axes, 4 hours

8.3 Shock

Parameter	Value
Half-sine	MIL-STD-810G, M
Peak Accelerati	20 g
Duration	11 ms

8.4 Humidity

Parameter	Value
Test	MIL-STD-810G, M
Conditions	85 degC / 85% R
Duration	1000 hours
Bias	Applied during

9. Drawing List

Drawing ID	Title	Size	Revision
XBDM-MEC-001	Module Outline	A3	A
XBDM-MEC-002	Mounting Holes	A3	A
XBDM-MEC-003	Connector Layou	A3	A
XBDM-MEC-004	PCB Stackup	A4	A
XBDM-MEC-005	Thermal Design	A3	A
XBDM-MEC-006	3D Model	-	A

10. Revision History

Version	Date	Author	Description
1.0	2026-09-09	Mechanical Engi	Initial release
